

Table Creation (DDL Operations)

Receptionist Table Creation

User: HOSPITALMGT

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```
CREATE TABLE Receptionist( R_Id NUMBER(4) Primary key,R_name VARCHAR2(20),R_shift VARCHAR2(10),
R_phone NUMBER(6),R_email VARCHAR2(20))
describe Receptionist
```

Object Type: **TABLE** Object: **RECEPTIONIST**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
RECEPTIONIST	R_ID	Number	-	4	0	1	-	-	-
	R_NAME	Varchar2	20	-	-	-	✓	-	-
	R_SHIFT	Varchar2	10	-	-	-	✓	-	-
	R_PHONE	Number	-	6	0	-	✓	-	-
	R_EMAIL	Varchar2	20	-	-	-	✓	-	-

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Fig 1: Description of the Receptionist Table

Maintain Table Creation

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```
Create table Maintain (Re_no number(5) Primary key, Re_date date, P_details varchar(8), R_Id number(4),  
constraint r foreign key (R_Id) references Receptionist (R_Id))  
  
describe Maintain
```

Object Type **TABLE** Object **MAINTAIN**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
MAINTAIN	RE_NO	Number	-	5	0	1	-	-	-
	RE_DATE	Date	7	-	-	-	✓	-	-
	P_DATAILS	Varchar2	8	-	-	-	✓	-	-
	R_ID	Number	-	4	0	-	✓	-	-
1 - 4									

Fig 2: Description of the Maintain Table

Staff Table Creation

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```
CREATE TABLE Staff (S_Id number(4) Primary key,S_name varchar(10),S_dept varchar(8),S_shift varchar(10))  
  
describe staff
```

Object Type **TABLE** Object **STAFF**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
STAFF	S_ID	Number	-	4	0	1	-	-	-
	S_NAME	Varchar2	10	-	-	-	✓	-	-
	S_DEPT	Varchar2	8	-	-	-	✓	-	-
	S_SHIFT	Varchar2	10	-	-	-	✓	-	-
1 - 4									

Fig 3: Description of the Staff Table

Patient Table Creation

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```
Create Table Patient (P_Id number(8) Primary Key, P_name varchar2(10), Age number(3), Gender varchar2(6),
P_phone number(8), Blood_Group varchar2(5))
describe patient
```

Object Type **TABLE** Object **PATIENT**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
PATIENT	P_ID	Number	-	8	0	1	-	-	-
	P_NAME	Varchar2	10	-	-	-	✓	-	-
	AGE	Number	-	3	0	-	✓	-	-
	GENDER	Varchar2	6	-	-	-	✓	-	-
	P_PHONE	Number	-	8	0	-	✓	-	-
	BLOOD_GROUP	Varchar2	5	-	-	-	✓	-	-
									1 - 6

Fig 4: Description of the Patient Table

Doctor Table Creation

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```
Create table Doctor (D_Id number (5) Primary key, D_name varchar(8), Li_no number(8))
describe Doctor
```

Object Type **TABLE** Object **DOCTOR**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
DOCTOR	D_ID	Number	-	5	0	1	-	-	-
	D_NAME	Varchar2	8	-	-	-	✓	-	-
	LI_NO	Number	-	8	0	-	✓	-	-
									1 - 3

Fig 5: Description of the Doctor Table

Serve Table Creation

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Create table Serve (P_Id number(8) Primary key, S_Id number(4), constraint s foreign key (S_Id) references staff (S_Id))

describe serve

Object Type TABLE Object SERVE

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
SERVE	P_ID	Number	-	8	0	1	-	-	-
	S_ID	Number	-	4	0	-	✓	-	-
									1-2

Fig 6: Description of the Serve Table

Consultation Table Creation

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Create table Consultation (D_Id number(5) Primary key, P_Id number(8), constraint p foreign key (P_Id) references patient (p_Id))

describe consultation

Object Type TABLE Object CONSULTATION

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
CONSULTATION	D_ID	Number	-	5	0	1	-	-	-
	P_ID	Number	-	8	0	-	✓	-	-
									1-2

Fig 7: Description of the Consultation Table

Room Table Creation

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Create table Room (R_no number(3) Primary key, R_type varchar(10), Rate number(5), Status varchar(15))

describe Room

Object Type: **TABLE** Object: **ROOM**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
ROOM	R_NO	Number	-	3	0	1	-	-	-
	R_TYPE	Varchar2	10	-	-	-	✓	-	-
	RATE	Number	-	5	0	-	✓	-	-
	STATUS	Varchar2	15	-	-	-	✓	-	-
									1 - 4

Fig 8: Description of the Room Table

Assign Table Creation

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Create table Assign (P_Id number(8) Primary key, P_name varchar2(10), Age number(3), Gender varchar2(6), P_phone number(8), Blood Group varchar2(5), R_no number(3), constraint no foreign key (R_no) references Room (R_no))

describe Assign

Object Type: **TABLE** Object: **ASSIGN**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
ASSIGN	P_ID	Number	-	8	0	1	-	-	-
	P_NAME	Varchar2	10	-	-	-	✓	-	-
	AGE	Number	-	3	0	-	✓	-	-
	GENDER	Varchar2	6	-	-	-	✓	-	-
	P_PHONE	Number	-	8	0	-	✓	-	-
	BLOOD_GROUP	Varchar2	5	-	-	-	✓	-	-
	R_NO	Number	-	3	0	-	✓	-	-
									1 - 7

Fig 9: Description of the Assign Table

Payment Table Creation

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```
create table Payment ( B_id number(6) primary key, A_due varchar2(6), B_date date, P_method varchar2(10),  
P_Id number(8),constraint pa foreign key (P_Id) references patient (p_Id))  
describe payment
```

Object Type **TABLE** Object **PAYMENT**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
PAYMENT	B_ID	Number	-	6	0	1	-	-	-
	A_DUE	Varchar2	6	-	-	-	✓	-	-
	B_DATE	Date	7	-	-	-	✓	-	-
	P_METHOD	Varchar2	10	-	-	-	✓	-	-
	P_ID	Number	-	8	0	-	✓	-	-

1 - 5

Fig 10: Description of the Payment Table

CEO Table Creation:

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```
create table CEO ( C_Id number(8) primary key, C_name varchar2(20), C_phone varchar2(11))  
describe ceo
```

Object Type **TABLE** Object **CEO**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
CEO	C_ID	Number	-	8	0	1	-	-	-
	C_NAME	Varchar2	20	-	-	-	✓	-	-
	C_PHONE	Varchar2	11	-	-	-	✓	-	-

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Fig 11: Description of the CEO Table

Enquire Table Creation

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```

CREATE TABLE Enquire (R_Id NUMBER(4) PRIMARY KEY, R_Name VARCHAR2(20), R_Shift VARCHAR2(10), R_Phone NUMBER(6), R_Email VARCHAR2(20),
S_Id NUMBER(4), S_Name VARCHAR2(10), S_Dept VARCHAR2(8), S_Shift VARCHAR2(10),
C_Name VARCHAR2(20), C_Phone VARCHAR2(11), C_Id NUMBER(8), CONSTRAINT sid FOREIGN KEY (S_Id) REFERENCES Staff (S_Id),
CONSTRAINT cid FOREIGN KEY (C_Id) REFERENCES CEO (C_Id))

```

[Describe enquire](#)

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
ENQUIRE	R_ID	Number	-	4	0	1	-	-	-
	R_NAME	Varchar2	20	-	-	-	✓	-	-
	R_SHIFT	Varchar2	10	-	-	-	✓	-	-
	R_PHONE	Number	-	6	0	-	✓	-	-
	R_EMAIL	Varchar2	20	-	-	-	✓	-	-
	S_ID	Number	-	4	0	-	✓	-	-
	S_NAME	Varchar2	10	-	-	-	✓	-	-
	S_DEPT	Varchar2	8	-	-	-	✓	-	-
	S_SHIFT	Varchar2	10	-	-	-	✓	-	-
	C_NAME	Varchar2	20	-	-	-	✓	-	-
	C_PHONE	Varchar2	11	-	-	-	✓	-	-
	C_ID	Number	-	8	0	-	✓	-	-

Fig 12: Description of the Enquire Table

Inserted Values in the tables

Receptionist:

R_ID	R_NAME	R_SHIFT	R_PHONE	R_EMAIL
7	Sadia	Day	17542	sadia@gmail.com
8	Hafsa	Day	16542	hafsa@gmail.com
10	Tahsin	Night	19542	tahsin@gmail.com

Maintain:

RE_NO	RE_DATE	P_DATAILS	R_ID
117	11-JUL-24	Anemia	7
118	29-MAR-24	Cancer	8
119	15-SEP-24	insomnia	10

Staff:

S_ID	S_NAME	S_DEPT	S_SHIFT
1	Mim	cardio	Day
2	Mahadi	Dermato	Night
3	Saima	Neuro	Night

Patient:

P_ID	P_NAME	AGE	GENDER	P_PHONE	BLOOD_GROUP
5	Zaman	19	Male	617295	O+
6	Maha	45	Female	712095	AB+
7	Asif	29	Male	811095	B-

Serve:

P_ID	S_ID
5	1
6	2
7	3

Doctor:

D_ID	D_NAME	LI_NO
1	Nafis	1457
2	Sumi	2880
3	Rakib	3574

Consultation:

D_ID	P_ID
1	5
2	6
3	7

Room:

R_NO	R_TYPE	RATE	STATUS
103	Single	800	Available
201	Double	1600	Unavailable
301	ICU	4300	Available

Assign:

P_ID	P_NAME	AGE	GENDER	P_PHONE	BLOOD_GROUP	R_NO
5	Zaman	19	Male	617295	O+	103
6	Maha	45	Female	712095	AB+	201
7	Asif	29	Male	811095	B-	301

CEO:

C_ID	C_NAME	C_PHONE
111	Sajid	1625781234

Payment:

B_ID	A_DUE	B_DATE	P_METHOD	P_ID
82497	2000	11-MAY-24	Online	5
84392	800	24-MAY-24	Cash	6
86780	1000	30-MAY-24	Cash	7

Enquire:

R_ID	R_NAME	R_SHIFT	R_PHONE	R_EMAIL	S_ID	S_NAME	S_DEPT	S_SHIFT	C_ID
7	Sadia	Day	17542	sadia@gmail.com	1	Mim	Cardio	Day	111
8	Hafsa	Day	16542	hafsa@gmail.com	2	Mahadi	Dermato	Night	111
10	Tahsin	Night	19542	tahsin@gmail.com	3	Saima	Neuro	Night	111

Query Test in DB

Single Row Subquery:

→ Show the P_Name, P_ID and Age of patients who have an age greater than the patient named Zaman.

```
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Select P name, P Id, age
from patient
where age > (select age from patient where p_name= 'Zaman')
```

Fig 13: Command for Single Row Sub Query

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
PATIENT	P_ID	Number	-	6	0	1	-	-	-
	P_NAME	Varchar2	8	-	-	-	✓	-	-
	AGE	Number	-	3	0	-	✓	-	-
	GENDER	Varchar2	8	-	-	-	✓	-	-
	P_PHONE	Number	-	12	0	-	✓	-	-
	BLOOD_GROUP	Varchar2	4	-	-	-	✓	-	-
1 - 6									

Fig 14: Description of the Single Row Sub Query

P_NAME	P_ID	AGE
Maha	6	45
Asif	7	29

Fig 15: Output for Single Row Sub Query

Simple view:

→ Create a view named as pview where gender male will be shown over the columns p_ID,P_Name,P_Phone from patient table.

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```

create view pview as
select p_ID ,P_Name,P_Phone
from patient
where Gender='Male'|

```

Fig 16: Simple View creation command

Object Type **VIEW** Object **PVIEW**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
PVIEW	P_ID	Number	-	8	0	-	-	-	-
	P_NAME	Varchar2	10	-	-	-	✓	-	-
	P_PHONE	Number	-	8	0	-	✓	-	-
1 - 3									

Fig 17: Description of the simple View

P_ID	P_NAME	P_PHONE
5	Zaman	617295
7	Asif	811095

Fig 18: Result of the Simple view as a whole table

Outer Joining:

Create an outer join between the CEO and Enquire tables using the C_Id column, and display the S_ID and R_ID columns.

```
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SELECT C.C_NAME, e.r_id,e.s_id
from ceo c, enquire e
where c.c_id (+) = e.c_id
```

Fig 19: Command for Outer joining between CEO & Enquire table

C_NAME	R_ID	S_ID
Sajid	7	1
Sajid	8	2
Sajid	10	3

Fig 20: output for Outer joining

Description of a Successful DB connection

We need DB connection:

- First, I searched “Mysql connector java”
- Then, I choose the “Maven Repository”
- I Clicked on ‘8.0.28’ version
- I Clicked on ‘Jar’ from files to download.
- Searched “Xampp” and downloaded it
- Then, I run the ‘Xampp’ file in ‘Run as administrator’.
- Installed the file and selected the location in ‘C drive’
- After that, I opened the ‘Xampp’ file from C drive
- Clicked on the ‘xampp-control.exe’ and pressed the Start button from ‘Apache’ and ‘MySql’.
- Lastly, pressed from SQL the admin button and it will open phpMyAdmin.

I created a database named **hospitalm**. Inside this database, I created a table named **Doctor** with the columns **D_ID**, **D_Name**, and **LI_NO**. Again, Created **Patient** table with **P_name,P_ID,Gender**. Lastly, Inserted values both tables.

D_ID	D_Name	LI_No
1	Nafis	1457
2	Sumi	2880
3	Rakib	3574

Fig 21: Created Doctor Table in the mysql

P_name	P_Id	Gender
Zaman	5	Male
maha	6	Female
Asif	7	Male

Fig 22: Created Patient Table in the mysql

Java connection

Register Driver: Dynamically load the MySQL driver class using '**Class.forName()**' to register it with the DriverManager.

Connect the DB: Establish a connection to the MySQL database with '**DriverManager.getConnection()**' by providing the required URL and username (root) without a password.

Create Statement: Create a statement object with '**createStatement()**' to execute SQL queries.

Execute the Query: Use the statement object to execute various SQL queries and process results with a ResultSet.

Connection Close(): Close the database connection to free up resources and maintain proper application performance.

Now I need an IDE. I used IntelliJ IDEA as my IDE.

After writing the database connection code in my project, I incorporated the MySQL Java Connector JAR file (mysql-connector-java-8.0.28.jar) to facilitate communication between Java and MySQL databases.

The output will be:

```
C:\Users\HP\.jdk\openjdk-21.0.1\bin\java.exe
connected
D_ID1 D_Name=Nafis LI-No =1457
D_ID2 D_Name=Sumi LI-No =2880
D_ID3 D_Name=Rakib LI-No =3574
6
```

```
run:
connected
P_nameZaman P_id=3 Gender =Male
P_nameMaha P_id=6 Gender =Female
P_nameAsif P_id=7 Gender =Male
BUILD SUCCESSFUL (total time: 0 seconds)
```

Fig 22: Execution of the Data base table (doctor and patient) in console